REVIEW 1 DELIVERABLE

Vehicle Service Centre DBMS

System Architecture, Conceptual ER Modeling & Relational Database Design (Project 10)

Introduction & Project Scope

Project Overview

This project entails the end-to-end design, normalization, and technical implementation of a Relational Database Management System (RDBMS) tailored specifically for multi-brand vehicle service facilities.

Scope of Key Deliverables

• Centralized repository for customer profiles, vehicle diagnostics, and repair logs.
• Automated service booking schedules and real-time job-card tracking.
• Spare parts inventory integration with dynamic stock deduction.
• Transparent billing module powered by mechanic labor logs and parts usage.



Problem Statement Analysis

Operational Fragmentation & Manual Overhead: Traditional service centers rely heavily on paper job cards, isolated spreadsheets, and physical registers. This absence of centralized data hierarchy makes tracking repair progress nearly impossible, leading to misplaced history, unrecorded spare parts, and frequent customer communication gaps.

Resource Scheduling & Stock Inaccuracies: Without automated schedule management, workshop heads struggle to assign mechanics efficiently based on specialization and availability. Scheduling conflicts delay delivery times, while untracked spare parts inventory causes unexpected stockouts and invoicing errors.

Core System Objective: To resolve these inefficiencies, this project engineers a robust 3NF normalized relational database that enforces integrity constraints, eliminates data redundancy, guarantees inventory precision, and automates billing workflows.

Core System Objectives

Digitization & Structural Integrity

Architect a fully normalized relational schema (3NF) linking customers, vehicles, appointments, and repair tasks into a cohesive, centralized database.

Eliminate data anomalies and ensure strict referential integrity constraints across all operational entities.

Process Automation & Financial Precision

Implement triggers and check constraints to prevent mechanic schedule overlapping and prevent issuing out-of-stock inventory items.

Automate financial calculations for mechanic labor fees and spare part totals on final billing invoices.

System Users & Functional Roles

Service Advisor

Registers customer profiles, logs vehicle specs, schedules service bookings, and produces finalized customer invoices.

Inventory Controller

Manages spare parts catalog, adjusts unit pricing, monitors reorder thresholds, and issues parts against verified job cards.

Service Mechanic

Accesses daily repair assignments, logs active working hours, updates task checklists, and requests necessary replacement parts.

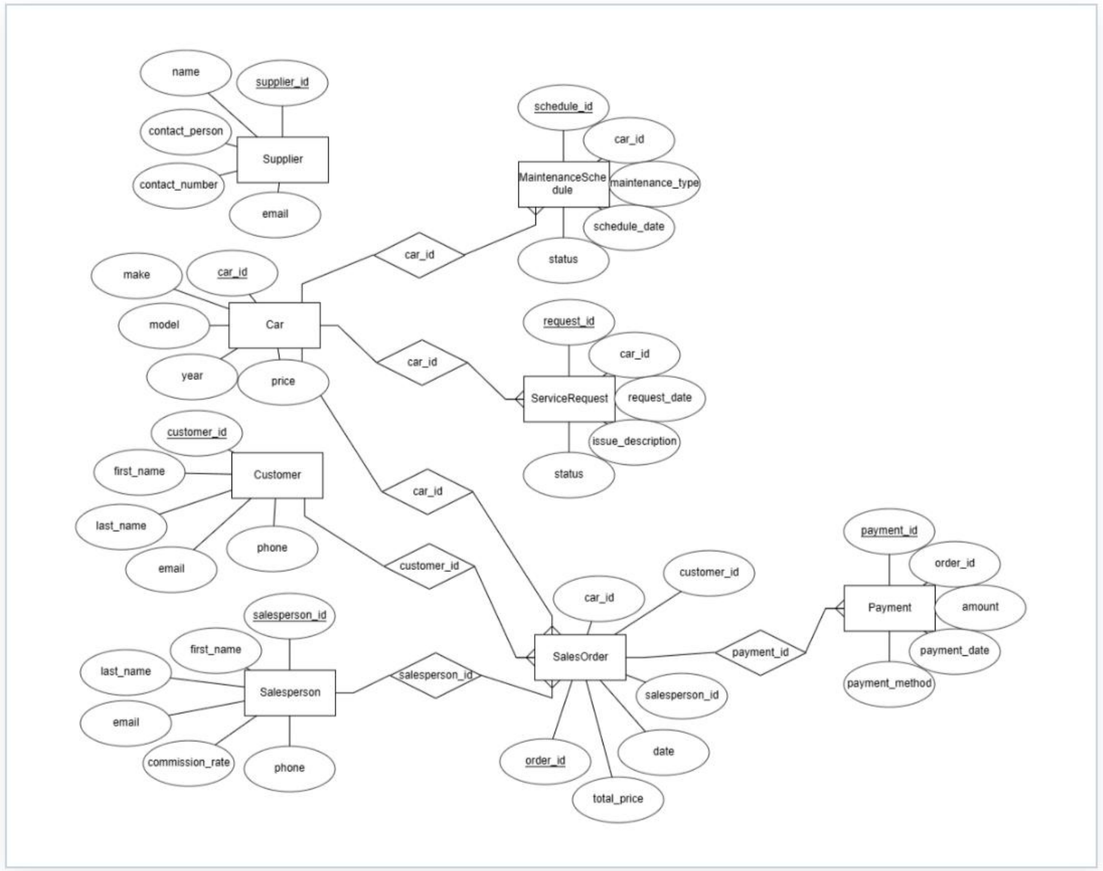
Workshop Manager

Monitors overall workshop productivity, evaluates mechanic performance, and reviews high-level revenue and service analytics.

Conceptual Design & ER Modeling

Formalizing entities, attributes, structural constraints, and relationship cardinalities.

Entity-Relationship (ER) Diagram



ER Model Structural Overview

Primary Entities: Customer, Vehicle, ServiceBooking, JobCard, Mechanic, ServiceTask, and SparePart.

Key Relationships & Cardinality:

- `Customer` to `Vehicle`: One-to-Many (1:N)
- `Vehicle` to `ServiceBooking`: One-to-Many (1:N)
- `ServiceBooking` to `JobCard`: One-to-One (1:1)
- `JobCard` to `Mechanic/Task`: Many-to-Many (M:N via Assignment)
- `JobCard` to `SparePart`: Many-to-Many (M:N via PartUsage)

Detailed Relationships & Constraints

1. Customer → Vehicle

Type: 1 : N (One-to-Many)
A customer can register multiple vehicles. Each vehicle belongs to exactly one primary customer (FK: `CustomerID` in Vehicle).

2. Vehicle → Booking → JobCard

Type: 1 : N and 1 : 1
A vehicle maintains multiple historical bookings. Each confirmed booking generates exactly one active `JobCard` record upon check-in.

3. JobCard ↔ Task ↔ Mechanic

Type: M : N Associative Entity
Resolved via `Assignment` table. Connects specific mechanics to repair tasks on a job card while tracking labor hours.

4. JobCard ↔ SparePart

Type: M : N Associative Entity
Resolved via `PartUsage` table. Tracks parts consumed per job card and locks historical unit prices for billing accuracy.

5. JobCard → Invoice

Type: 1 : 1 (One-to-One)
Each completed job card produces one invoice summarizing labor charges, parts subtotal, applied taxes, and payment status.

6. Key Business Constraints

Integrity Enforcements: CHECK constraints mandate non-negative inventory stock, valid status progressions, and strict primary key uniqueness.

Relational Schema Mapping (3NF)

RELATION / TABLE	PRIMARY KEY (PK)	FOREIGN KEYS (FK)	KEY ATTRIBUTES
CUSTOMER	`CustomerID`	None	Name, Phone, Email, Address
VEHICLE	`RegNo`	`CustomerID`	Make, Model, Year, FuelType
SERVICE_BOOKING	`BookingID`	`RegNo`	BookingDate, Status, SlotTime
JOB_CARD	`JobCardID`	`BookingID`	DateIn, DateOut, JobStatus
ASSIGNMENT	`AssignmentID`	`JobCardID`, `TaskID`, `MechanicID`	StartTime, EndTime, Status
PART_USAGE	`UsageID`	`JobCardID`, `PartID`	Quantity, BilledUnitPrice
INVOICE	`InvoiceID`	`JobCardID`	LaborTotal, PartsTotal, NetAmount

Review 1 Deliverable Complete

Conceptual ER Model, Relational Schema & 3NF Normalization Finalized.

Thank You — Questions & Feedback Welcome

Image Sources



https://www.roccoshonda.com/wp-content/uploads/al_opt_content/IMAGE/www.roccoshonda.com/wp-content/uploads/2026/01/car_diag.jpg.bv_resized_mobile.jpg.bv.webp?bv_host=www.roccoshonda.com

Source: www.roccoshonda.com



https://media.geeksforgeeks.org/wp-content/uploads/20240307113951/ER-Diagrams-for-Automobile-Dealership-Management.png

Source: www.geeksforgeeks.org